

Ex2 Introduction

27/2/2026

Released Thursday

Aims

- To test knowledge on LTR for information retrieval
 - Build upon ranking and evaluation experience from Exercise 1
 - Consider complex cascading ranking pipelines in IR
 - Deploy and evaluate learning-to-rank techniques
 - Experience challenges in feature engineering, for both query dependent and query independent features

Mechanics

- The Colab notebook downloads all the data that you need, and provide some skeleton code, in particular:
 1. A revised index for Exercise 2 with more **fields information** – (see Lecture 7) and more document metadata (title, **URL**, **body**)
 2. A baseline feature configuration involving weighting model on a field, and a QI feature
 3. Test cases used for assessing implementations
- There is a specification document for Ex2 – the aim here is to cutdown the length of the notebook

Techniques

- URL length features
 - See also Lecture 9
- Proximity feature
 - Detailed at length in the specification
 - See also Lecture 7 for theory
- LambdaMART LTR and feature importance
 - See also Lecture 8

Reference Material

- The Specification!
- Lectures 7, 8 & 9
- PyTerrier documentation:
 - Operators - <https://pyterrier.readthedocs.io/en/latest/operators.html>
 - Learning to Rank - <https://pyterrier.readthedocs.io/en/latest/ltr.html>
 - Running Experiments - <https://pyterrier.readthedocs.io/en/latest/experiments.html>
 - pt.apply Custom Transformers - <https://pyterrier.readthedocs.io/en/latest/apply.html>

Breakdown

- Q1 – Baseline Effectiveness – 6 marks
- Q2 – URL QI Features – 22 marks
- Q3 – Proximity Feature – 14 marks
- Q4 – Additional LTR Experiments – 8 marks

Builds on Exercise 1

Deadline **19st March 4.30pm**

Do give a read at the Exercise ASAP, and let us know questions

>> LAB NEXT WEEK– BUT we want you to get started ASAP <<

>> Do not delay <<

Final Advice

- IR Exercises are designed to need *thinking*
- You cannot complete an exercise in 1 sitting
- The coding is NOT enough
- The Ex2 index is larger – it will take a while to run training and evaluation – but we have made things faster through transformer caching
- Do NOT leave this exercise to the last minute
 - Start well before next week's lab